Topological organization of a stochastic stellar oscillation spectrum: an auditable Kepler–TESS case study of KIC 10963065

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Abstract

We present a reproducible, failure-aware benchmark of spectral persistence and cylindrical echelle geometry for the solar-like oscillator KIC 10963065 (Rudy). Twenty-one checksum-verified public Kepler and TESS products anchor a complete observation capsule. The original Lightkurve tutorial gives $\nu_{\max} = 2145\,\mu$Hz and $\Delta\nu = 103.1121\,\mu$Hz; a distinct Harvey-whitened envelope and ACF pipeline gives 2233.60 and $103.3637\,\mu$Hz. Localized finite path H_0 lifetimes equal elder-rule peak prominence exactly. Their weighted location, $2105.25\,\mu$Hz, is retained as a diagnostic, not a calibrated replacement for the classical envelope. We report window-matched conditional background tests, alias alternatives, 72 registered sensitivity diagnostics, monthly resampling and stochastic realization scatter. Cylindrical paths remain unidentified and do not establish estimator utility or new frequency-ratio measurements. Delay topology is omitted under an earlier registered negative decision. A 396-cell same-star degradation experiment uses all eleven real TESS sector windows; actual TESS candidates remain unconfirmed. The result is an auditable single-star case study with bounded claims, not population validation or evidence of new information beyond classical controls.

1 Introduction

Solar-like oscillations are stochastic: the spectrum depends on mode excitation, damping, finite observing duration, cadence integration and the spectral window. The usual global observables $\nu_{\max}$ and $\Delta\nu$ summarize an oscillation envelope and a nearly regular acoustic-mode comb. Echelle diagrams organize the comb and expose degree-dependent ridges. Their interpretation is inseparable from quality masks, background models and the distinction between one realization and a population of stars (Lund et al., 2017; Silva Aguirre et al., 2017).

Persistent homology tracks connected components and loops through a filtration. Applied to a one-dimensional superlevel spectrum, it provides localized births and merger saddles. This can be a compact organization of spectral peaks, but it does not by itself establish additional physical information or estimator accuracy. Known diagram-stability results concern perturbations of a fixed field and complex, not the correctness of a spacing selected from a collection of folds (Cohen-Steiner et al., 2007). A cylindrical echelle additionally requires a periodic phase boundary: the left and right display edges represent the same phase.

TopoAstero's development sequence separated analytic validation, stochastic simulations, classical controls and mathematical audits before this case study. The delay branch failed its registered simulation tests and was dropped. The cylindrical branch passed geometry tests but failed estimator-utility and interval coverage gates; its scientific scope remains geometry only. The present paper therefore asks whether a fully auditable observation pipeline can connect these constructions to a real star while

preserving their demonstrated limits. Rudy was already the descriptive benchmark in the classical pipeline. It is not an untouched confirmation star, and this study does not open population or final holdout partitions.

2 Observations and locked analysis capsule

2.1 Public products and provenance

The Kepler short-cadence benchmark comprises ten tutorial products across Q2, Q5, Q6 and Q7. The primary finite, quality-filtered series retains 430387 cadences over an absolute span of 488.18 days with cadence-mask duty 0.600. Eleven homogeneous SPOC two-minute TESS light curves cover sectors 14, 15, 40, 41, 53, 55, 74, 75, 80, 81 and 82 for TIC 48450369. Every product is identified by a frozen MAST URI, payload SHA256 and size; the capsule records flux, quality, time system, BJD reference, cadence and exposure metadata. A missing or changed payload stops execution rather than replacing the registered bytes. The official tutorial supplies the classical reproduction context (Hall and Barentsen, 2020; Lightkurve Collaboration et al., 2018; Mikulski Archive for Space Telescopes, 2026).

2.2 Preprocessing roles

PP-01 uses PDCSAP flux and the Lightkurve default Kepler quality mask, with only mission detrending and per-month normalization. PP-02 uses quality equal to zero and five-sigma clipping. PP-03 uses SAP flux, the default mask, five-sigma clipping and a 1001-cadence Savitzky–Golay high-pass. PP-04 uses the hardest quality mask and five-sigma clipping. PP-05 supplies analytic FFT/unit tests; PP-06 applies the default TESS quality mask to individual SPOC sectors. Nonfinite flux is removed before inference. Absolute timestamps and gaps are preserved. Integer quality masks and per-product cadence counts are recorded from the locked installation. No branch is chosen using a topological score.

The Stage 10 configuration was locked before its new topology and null results. This is an implementation lock, not an externally timestamped preregistration. Earlier Stage 3 descriptive Rudy results were already known. This distinction is part of the claim ledger and prevents the present results from being presented as an independent discovery.

3 Methods

3.1 Classical spectrum and background

We reproduce the Lightkurve PSD, default log-median flattening, ACF2D global parameters and uncorrected scaling quantities separately from the independent Harvey/ACF pipeline. The latter evaluates a one-sided Lomb–Scargle PSD from 100 to 8000 μHz in ppm^2/μHz, comparing zero-, one- and two-component rational Harvey continua with exponent four. The interval 1500–2800 μHz is excluded from background fitting. Minimum BIC chooses the primary continuum; all families remain in the sensitivity table. Because windowed ordinates can be correlated, BIC is a descriptive ranking, not a posterior probability.

Native ordinates are mean-rebinned to 0.2-μHz centers for the reduced spectrum. When a diagnostic output grid is finer than a segment's native resolution, interpolation is confined to supported frequencies and is explicitly not counted as additional information. The primary classical envelope maximizes a 35-μHz Gaussian-smoothed P/B in 1700–2500 μHz. The spacing estimator uses a 0.5-μHz smoothed whitened spectrum in a 275-μHz half-width, with an ACF search around the classical scaling prediction.

This is a distinct estimator from the tutorial. PSD units and FFT endpoint conventions are independently tested (VanderPlas, 2018; Astropy Collaboration et al., 2022).

3.2 Localized spectral persistence and alias controls

The primary filtration is exact superlevel path H_0 on $f(\nu) = \log[P(\nu)/B(\nu)]$, after 1-$\mu$Hz smoothing of power. For a finite component born at a peak and merged at a saddle, $\pi = f_{\rm birth} - f_{\rm merge}$. The elder rule makes π exactly the corresponding elder-rule prominence; an independent implementation checks the identity on the observation. Essential components retain localization but have no finite lifetime and are excluded from numerical summaries. Ratio, residual and a z-score using a registered 10% background allowance are mandatory alternative normalizations; the allowance is not a fitted posterior.

The persistence-weighted centroid in 1700–2500 μHz is compared with the classical envelope. A pair-difference/autocorrelation comb diagnostic scans 57 spacings from 20 to 300 μHz. ACF and power-spectrum-of-power-spectrum (PSPS) controls use the same supported spectrum. Boundary winners, half/double alternatives and disagreement are retained. Neither the centroid nor the broad comb maximum is automatically interpreted as a calibrated seismic estimate.

3.3 Periodic echelle geometry

We fold the log-ratio spectrum to a 24-phase-bin finite cylinder, with phase periodic and the row direction finite. Orders lacking 75% phase support contribute no vertices. The frozen Stage 8 score combines normalized finite H_0 persistence and a topology-free circular ridge graph:

$$S_{\rm cyl} = 0.55\, S_{H_0} + 0.45\, S_{\rm graph}.$$

The registered classical ACF anchor determines half, nominal and double families, each evaluated at relative spacings 0.90, 0.95, 1.00, 1.05 and 1.10. The published spacing is never an anchor for score selection. Each null realization repeats the classical-anchor policy. Graph, conventional folded-profile contrast, Hough, ACF and PSPS controls accompany the topology.

A periodic loop can encode the geometry of the domain without establishing a stellar ridge identity. The competing graph paths have no defensible angular-degree anchor. Accordingly we report no inferred r_{02}, r_{01} or r_{10}. Ratios computed from externally identified published modes are formula audits, not topological measurements. The new conditional ridge proposition from Stage 9 is still awaiting independent proof review and is excluded from headline claims.

3.4 Conditional nulls and finite Monte Carlo precision

We draw 256 stationary Gaussian background realizations from the selected Harvey continuum on a full Kepler cadence grid spanning the real absolute times. The generative density is corrected by measured sampling duty to reconcile the full-span LS density with retained-exposure white-noise variance. Draws are corrected for the analytic mean two-tap linear-interpolation power response, interpolated to the observed barycentric timestamps and processed through the same native LS frequency policy, rebinning and smoothing. This discretization approximates the finite sampling operator; the ensemble mean whitened power, 1.002, is retained as a nuisance-model diagnostic.

For each statistic family we retain the maximum over its registered frequency/spacing search, giving

$$p = \frac{1 + \#\{S^{(b)}_{\rm max} \geq S_{\rm obs}\}}{257}.$$

Upper-tail ties are counted conservatively. Wilson intervals describe the underlying simulated exceedance probability; MC standard errors and counts are reported. Holm correction controls the seven reported families within this conditional model. The minimum attainable unadjusted p is $1/257$. The fit is not reestimated in each draw, and the test does not calibrate background-fit uncertainty, instrumental residuals or nonlinear detrending. No raw-bin Gamma bound is applied to a smoothed or selected score.

3.5 Uncertainty and stellar scaling

Quarters and cumulative absolute baselines are analyzed with the full-span background held fixed, after converting each segment PSD by segment duty divided by full-span duty to the same density convention. Cumulative points overlap. Ten nonoverlapping monthly PSDs receive the same duty conversion, are resampled 256 times and averaged by retained cadence count. These intervals target the monthly-averaged-spectrum estimator, which differs from the coherent full-span periodogram. Sixty-four Gaussian draws of a smoothed observed PSD describe conditional stochastic repeat scatter. Sixty-four monthly cylindrical resamples provide alias diagnostics only, because coverage failed the earlier simulation gate.

We retain these components alongside the range across all 72 registered branch diagnostics. They are not combined as independent errors. Uncorrected scaling calculus uses $T_{\rm eff} = 6140 \pm 77$ K and $M \propto \nu_{\rm max}^3 \Delta\nu^{-4} T_{\rm eff}^{3/2}$, $R \propto \nu_{\rm max} \Delta\nu^{-2} T_{\rm eff}^{1/2}$ and $g \propto \nu_{\rm max} T_{\rm eff}^{1/2}$. Draws preserve paired seismic bootstrap values and separate temperature variation. Composition, structural corrections and solar-reference systematics remain unquantified. The scaling outputs are conditional diagnostics, not a physical stellar posterior (Viani et al., 2017; Silva Aguirre et al., 2017).

3.6 Kepler-to-TESS degradation

We integrate the piecewise-linear Q5 Kepler curve through ideal 120-second boxcar exposures at source offsets 0, 30 and 60 days. Exposures crossing source gaps longer than 180 seconds are rejected. The experiment applies all eleven actual TESS retained-timestamp windows, duration cuts of 13.7 and 27.4 days, additional retention fractions 1 and 0.8, and added-noise variance factors 0, 1 and 2. Additional white noise matches the positive difference between the actual TESS and degraded Kepler high-frequency noise floors; existing Kepler noise is never subtracted.

The 396 cells record actual duration, duty and source-coverage loss. They reuse correlated source realizations and do not represent independent stars. Actual sectors are evaluated separately using the earlier fixed feasibility rule (envelope SNR at least 1.20, reference-conditioned envelope/spacing agreement and ACF peak at least 0.08). The native 1500–2700-μHz sidebands, 1800–2500-μHz search and primary tutorial reference values exactly preserve the Stage 3 feasibility policy. This rule yields marginal candidates, not blind detections or confirmations. The ideal boxcar cannot match TESS bandpass, detector systematics, detailed exposure dead time or nonlinear cotrending.

4 Simulation evidence and inherited decisions

The registered Stage 4 simulation design spans observing duration, cadence, duty, noise, mode lifetime, visible orders, background mismatch and alias conditions. Stage 6 showed exact prominence equivalence and supported an equivalence-limited spectral claim, with explicit null and classical controls. The Stage 7 evaluation yielded zero BH rejections for both phase-randomized and IAAFT-surrogate paths and no positive held-out predictive increment; delay PH was dropped from the critical path.

Stage 8 had only five detectable evaluation cells. Correct-alias accuracy was 0.20 for the cylindrical

score and 0.60 for the graph control; nominal 90% interval coverage was 0.20. Thus geometry passed and estimator utility did not. These are small simulation-gate summaries, not population estimates. The current Rudy result cannot reverse them merely by agreeing with a published spacing. Figure 7 locates the observations against the simulation coordinates without treating the comparison as inverse calibration.

5 Rudy results and ablations

5.1 Classical baseline and spectral organization

The tutorial is reproduced at 2145 and 103.1121 μHz. The independent primary envelope and spacing are 2233.60 and 103.3637 μHz. Differences reflect pipeline conventions and finite realization behavior, rather than a correction forced to a reference. The published LEGACY context is $\nu_{\max} = 2203.7^{+6.7}_{-6.3}\,\mu$Hz and $\Delta\nu = 103.179 \pm 0.027\,\mu$Hz (Lund et al., 2017). Published-mode overlays inspect local spectral consistency; selecting a peak near an externally supplied frequency is not blind peak bagging. Table 1 retains all preprocessing branches.

The primary persistence centroid is 2105.25 μHz, displaced from the classical envelope. Finite lifetimes reproduce prominence to numerical precision. This provides localized organization without establishing a new signal-strength observable. The broad comb diagnostic has a boundary maximum and competing aliases, while ACF/PSPS alternatives differ. The cylinder's largest registered score occurs in the double family at 217.06 μHz. We retain that choice and all competing trials as a diagnostic rather than declaring a new spacing measurement.

5.2 Conditional tests and sensitivity

Table 2 gives all seven maximum-statistic tails and Holm-adjusted values. Rejection, where present, is evidence against the fitted conditional background-only model. It does not demonstrate topology's incremental utility relative to the graph, ACF, PSPS or prominence controls. The 48 preprocessing–background–normalization combinations and 24 one-factor background-scale, smoothing and resolution diagnostics remain visible. Background families can change envelope contrast and centroid interpretation even when a classical comb spacing is comparatively stable. No favorable branch is selected after examining the score.

Table 3 separates the monthly estimator, full-span point, monthly bootstrap interval and pipeline range. Quarter and cumulative diagnostics show observed window/realization dependence. Conditional stochastic scatter and seismic/temperature scaling tables are supplied separately. Pipeline ranges and realization scatter are not independent stellar error budgets.

5.3 Transfer results

The degraded Kepler series is evaluated through every registered real TESS window, and actual sectors occupy the same noise/duration context. Under the fixed primary feasibility policy, marginal actual-sector candidates occur in sectors 15. They remain unconfirmed. The empirical conditional fractions in Figure 10 describe the reused Q5 experiment, not a binomial rate for independent realizations or a population detectability law.

Table 1: Mandatory preprocessing branches. The tutorial and independent estimators use different whitening and envelope conventions. All frequencies are in μHz.

Branch	Cadences	Tutorial $\nu_{\max}$	Tutorial $\Delta\nu$	Classical $\nu_{\max}$	Classical $\Delta\nu$
PP-01	430387	2145	103.1121	2233.6	103.3637
PP-02	363690	2175	102.9461	2152.8	103.2427
PP-03	430158	2175	102.5431	2152.2	103.2803
PP-04	363690	2175	102.9461	2152.8	103.2427

Table 2: Conditional fitted-background maximum-statistic tests. MC uncertainty and exceedance counts are provided in the machine-readable table. Holm controls the seven reported families within this conditional model.

Statistic family	Observed	Plus-one p	Holm p
Smoothed envelope	1.4326	0.0039	0.0272
Largest finite peak lifetime	2.1213	0.0039	0.0272
Persistence comb maximum	0.0757	0.0039	0.0272
Cylindrical combined maximum	0.3893	0.0039	0.0272
Ridge graph maximum	0.5861	0.0039	0.0272
ACF maximum	0.6781	0.0039	0.0272
PSPS maximum	0.0537	0.0039	0.0272

Table 3: Distinct uncertainty targets. Bootstrap endpoints target the monthly averaged spectrum. The full-span point is a different estimator. Pipeline ranges include all 72 registered diagnostics; components are not combined.

Quantity (μHz)	Full span	Monthly point	Monthly bootstrap 95%	Pipeline range
Classical envelope	2233.60	2232.60	[2142.10, 2435.70]	[2146.40, 2236.30]
Classical spacing	103.36	103.42	[102.89, 103.57]	[103.23, 103.37]
Persistence centroid	2105.25	2098.05	[2087.14, 2112.25]	[2092.48, 2125.69]

6 Discussion and limitations

The useful deliverable is a fully inspectable connection between observations, classical diagnostics, localized persistence, periodic geometry and failure reasons. Known path-prominence equivalence prevents a novelty claim based solely on peak lifetime. A periodic complex avoids a geometric seam artifact, but a correct domain topology does not validate an estimator or assign degree labels. The previous simulation failures remain part of the result.

The reference star was already inspected; nulls condition on a fitted stationary continuum; Gaussian predictive draws do not reconstruct the complete excitation and damping physics; quarter and source-window experiments share data. Background-fit uncertainty, nonstationary instrument effects and stellar scaling corrections remain outside the calibrated model. The finite null ensemble cannot support extremely small false-alarm probabilities. Topological ratios and population-level conclusions are withheld. The accompanying limitation register records these restrictions claim by claim.

Independent human proof review of the new Stage 9 proposition and independent Stage 10 reproduction/claim review remain pending. The current artifact is a complete internal review draft and code freeze. Automated clean-process reproduction is recorded separately; it is not an independent scientist's attestation. This distinction protects the manuscript from overstating either scientific validation or publication status.

7 Data, code and reproducibility availability

TopoAstero 0.11.0 supplies the frozen public identifiers, hash-verified acquisition workflow, locked Python environment, configuration, per-cadence-count provenance, all numerical tables, eight-claim ledger and ten PDF/SVG/PNG figure sets. Raw public FITS files are reacquired into an external cache and are not bundled. The command documented in the repository, `uv run python workflows/reproduce_stage10.py`, with a cache argument and `-download`, regenerates the analysis and draft from public identifiers. A fresh source copy, locked environment and new process check numeric tables and headline outputs. Reduced-table replay is available for offline CI and is labeled separately from raw-product reproduction.

The archive includes this editable LaTeX source, generated tables, checked bibliography and compiled draft. No public code repository, journal submission, Zenodo deposition or DOI is represented as completed. Those publication identifiers must be supplied by the authors after the corresponding external actions. The internal snapshot lists exact file hashes and the pending independent review record.

8 Conclusions

Rudy connects a reproducible classical baseline to localized spectral persistence and a correctly periodic echelle construction. The spectral representation reorganizes known prominence information; cylindrical scores and unidentified ridges remain geometry diagnostics under the inherited gate. Explicit null scope, aliases, branch sensitivity and distinct uncertainty estimands narrow the admissible claims. The Kepler–TESS experiment demonstrates a reproducible same-star transfer protocol, with actual TESS observations retained as context. No new degree-blind ratios, calibrated topological stellar parameters or population generalization follows from this case study.

A Journal figure suite

Every figure uses TopoAstero's unmodified academic style TA-PLOT-1.0 and lists its source tables in the machine-readable figure manifest.

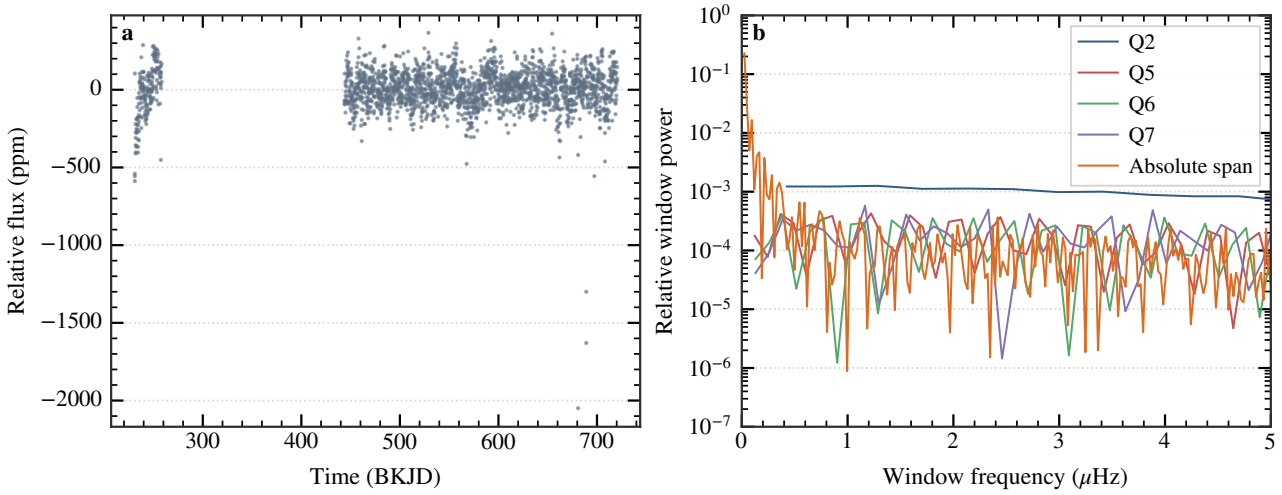


Figure 1: Absolute-time Kepler observations (every 200th cadence for display) and cadence-mask windows. Inter-quarter gaps are retained.

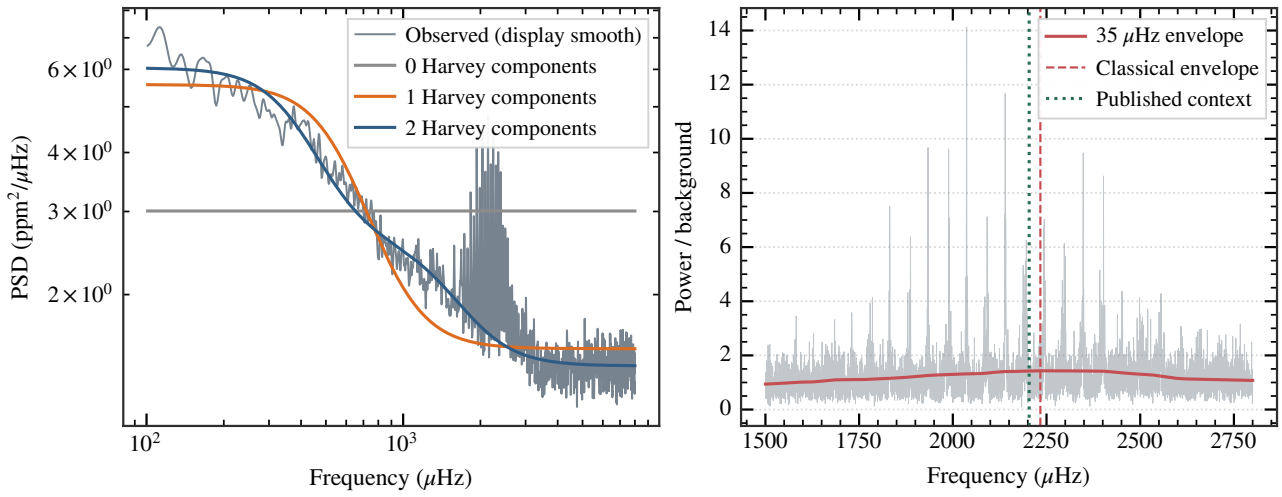


Figure 2: PSD, all registered background families, and the independently estimated envelope. The published context is not a tuning target.

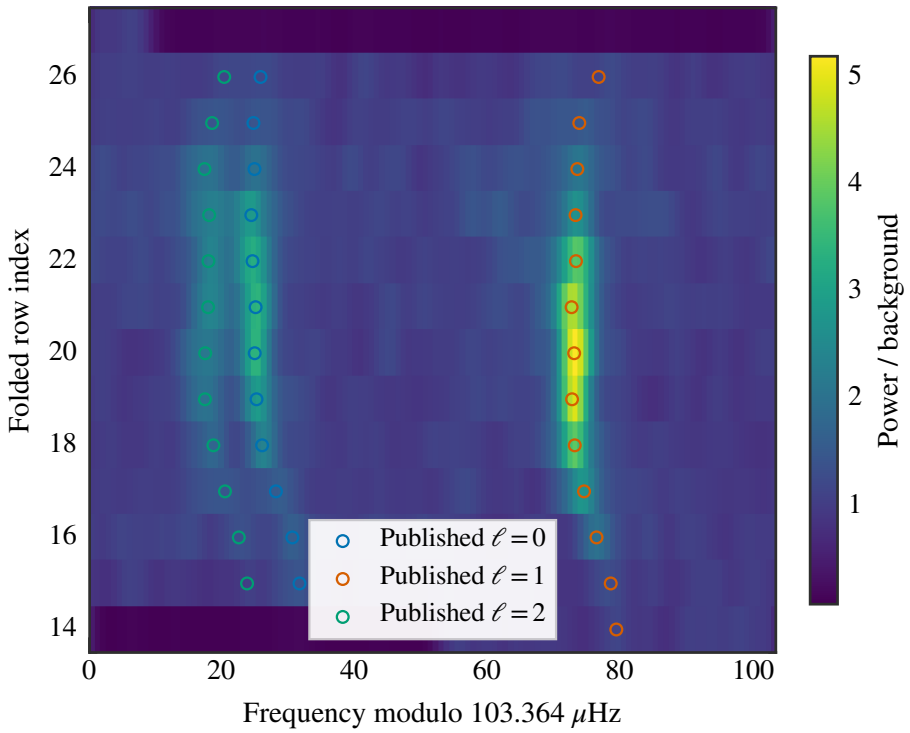


Figure 3: Ordinary echelle at the classical ACF spacing, with external degree-identified modes. Folded rows are not physical radial-order labels.

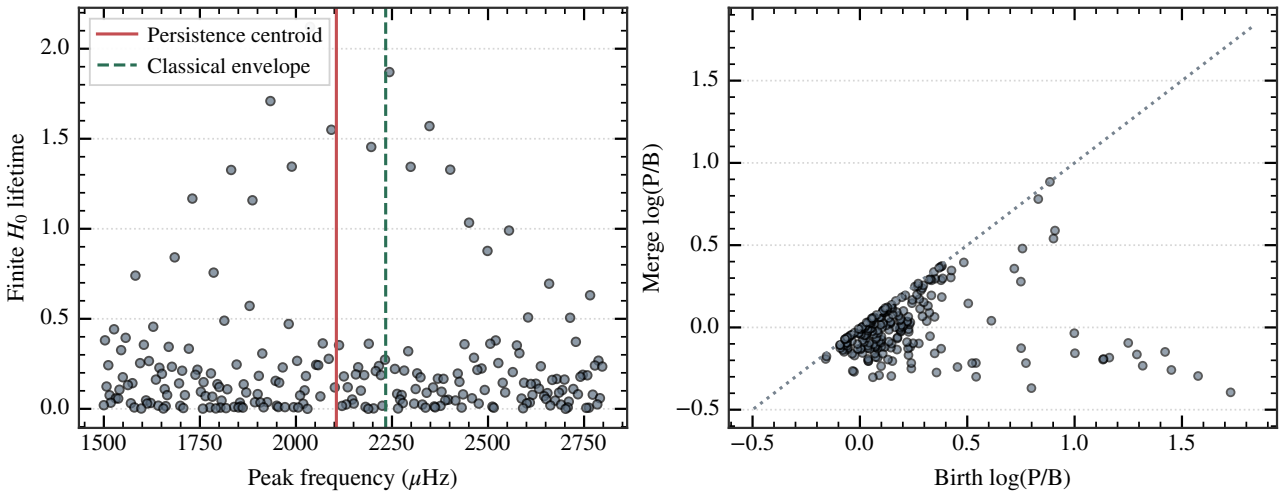


Figure 4: Finite peak lifetimes and the localized birth–merge diagram. Essential classes are kept in JSON but excluded from these numeric plots.

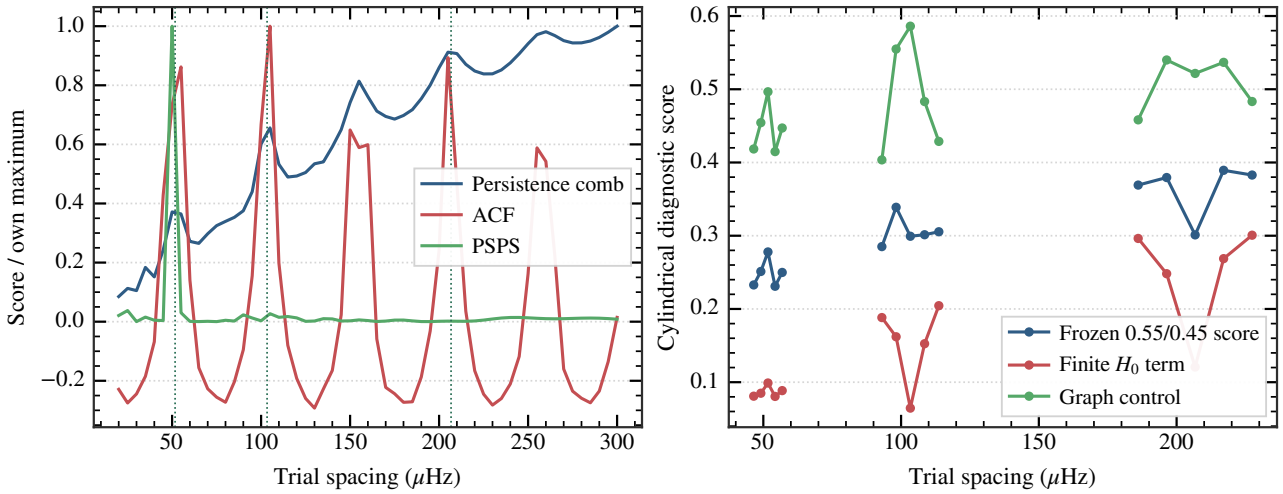


Figure 5: Spectral and cylindrical scans retain boundary, half, nominal and double alternatives. Score selection is not calibrated estimator utility.

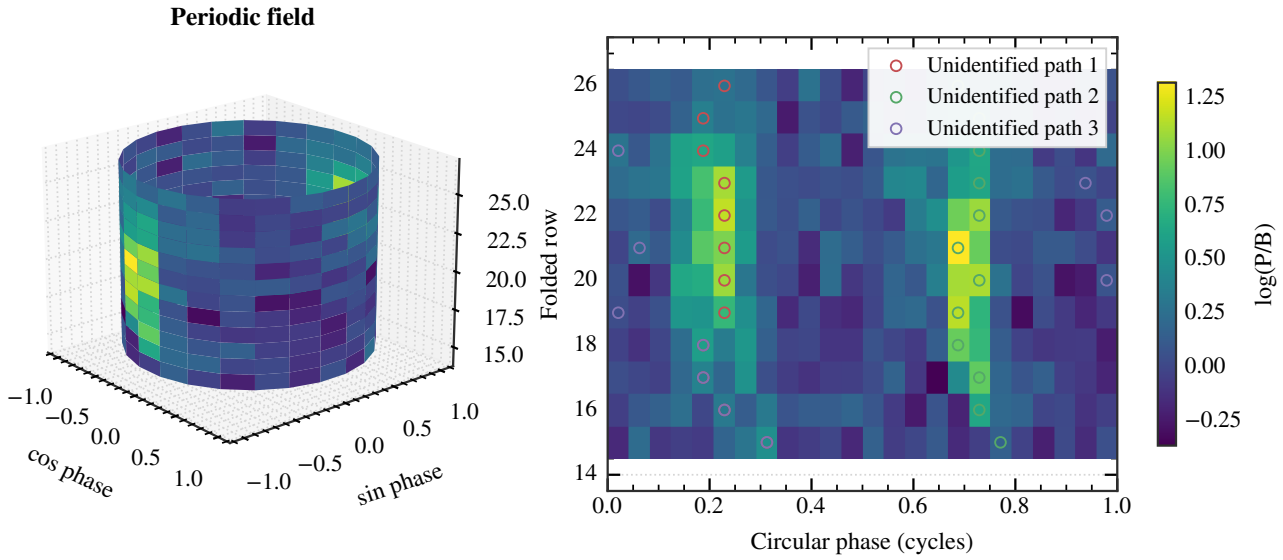


Figure 6: Periodic field and competing unidentified circular graph paths. No topological frequency ratio is inferred without a defensible degree anchor.

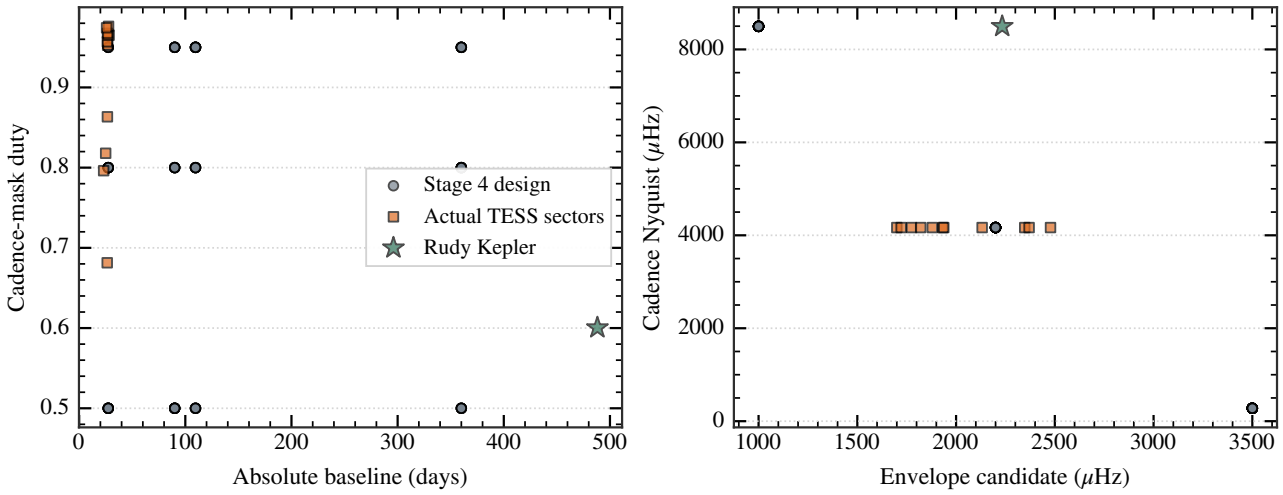


Figure 7: Observed coordinates against the registered simulation design. TESS envelope candidates do not imply detections; this is domain context, not inverse simulation calibration.

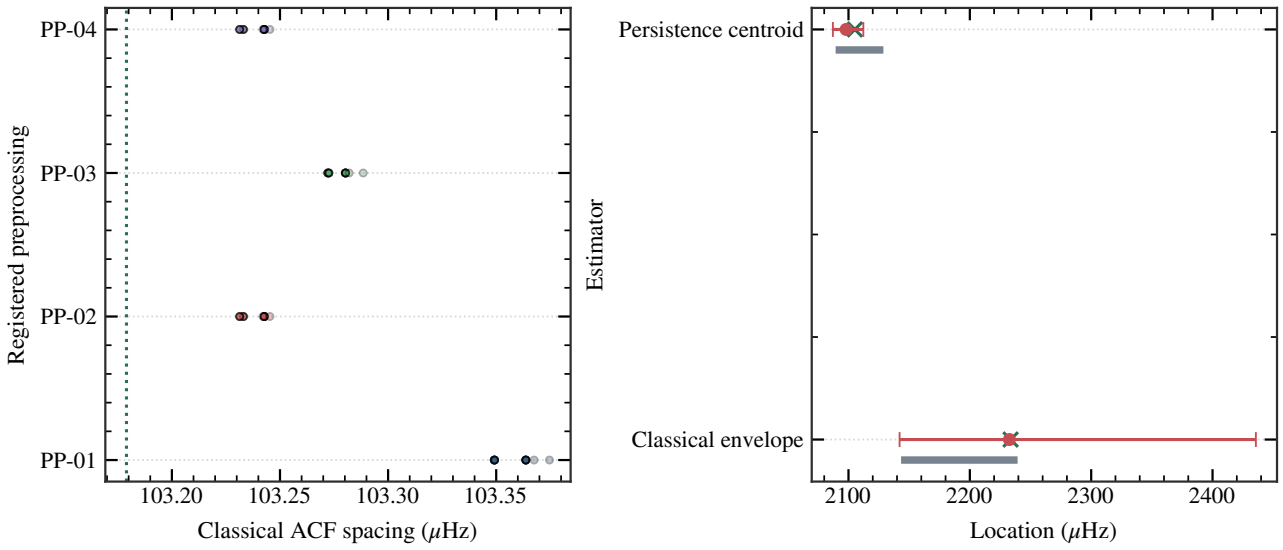


Figure 8: All registered sensitivity diagnostics and distinct uncertainty targets. Monthly bootstrap bars, pipeline ranges and full-span points are not combined as independent error budgets.

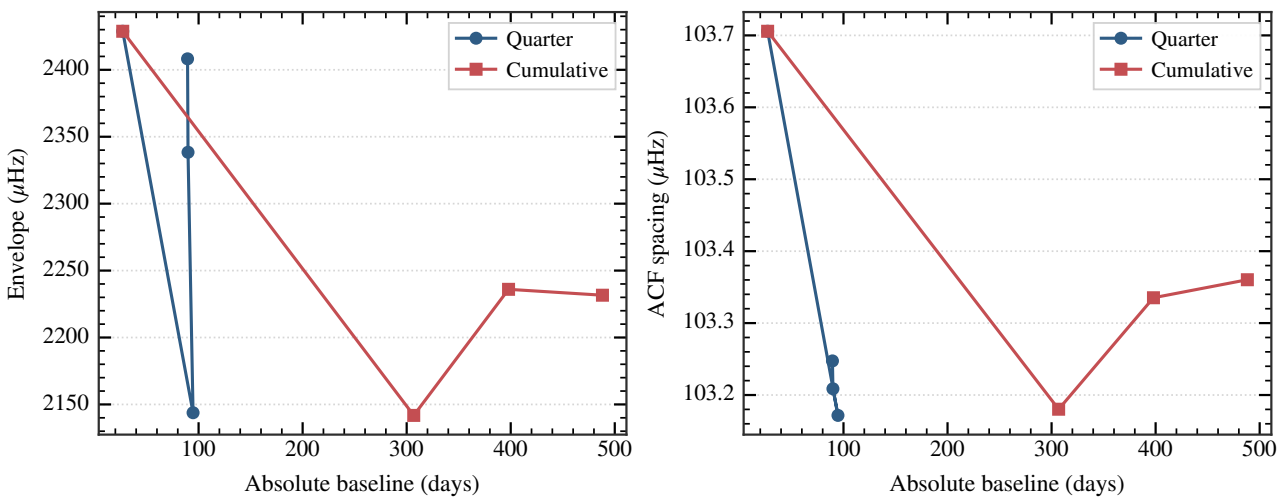


Figure 9: Quarter and overlapping cumulative-baseline diagnostics with the primary background held fixed.

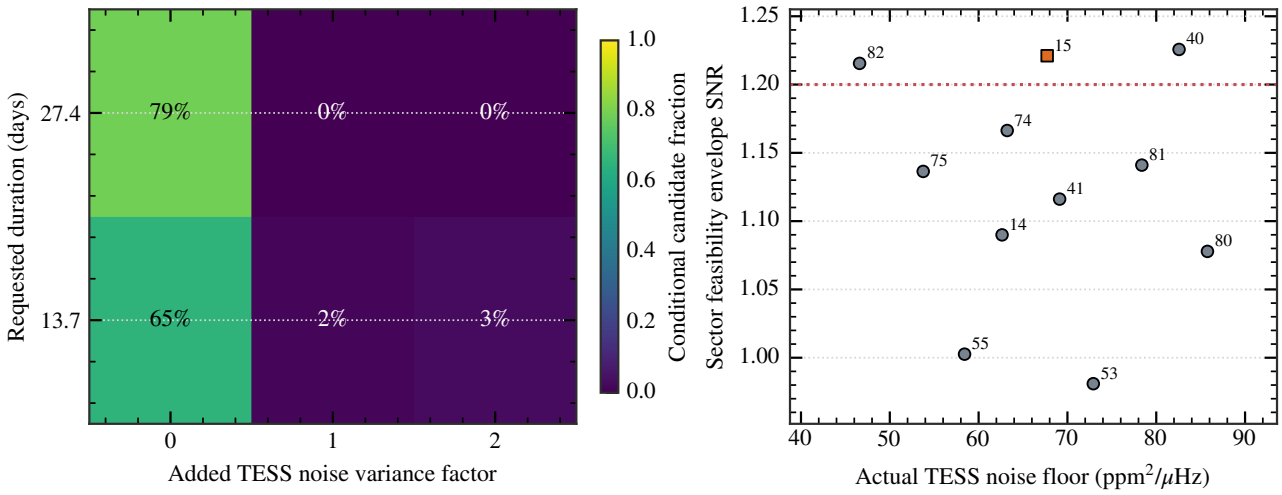


Figure 10: Conditional Kepler degradation fractions and actual TESS sector context. Reused Kepler offsets and real windows are correlated; marginal candidates are not confirmations.

Frozen claim–evidence ledger

P1-C01 The public-product capsule and tutorial baseline are reproducible. *Control:* Lightkurve baseline, frozen payload SHA256. *Limit:* single star and locked pipeline.

P1-C02 Finite path H0 lifetimes are elder-rule prominence, with frequency localization. *Control:* independent path prominence. *Limit:* no incremental information or novelty claim.

P1-C03 Conditional fitted-background null results are reported with full search maxima and Holm correction. *Control:* window matched stationary Gaussian background, ACF, PSPS, graph. *Limit:* background fit uncertainty and instrument systematics not calibrated.

P1-C04 Rudy comb and cylindrical scores remain ambiguous diagnostics. *Control:* ACF, PSPS, conventional echelle, Hough, ridge graph. *Limit:* G8 geometry only no utility or validated coverage.

P1-C05 Monthly resampling, stochastic scatter and registered pipeline ranges target different uncertainties. *Control:* quarter and cumulative baselines, 72 registered branches. *Limit:* no combined stellar posterior no population inference.

P1-C06 TESS degradation is tested with all real sector windows and actual sectors retained as context. *Control:* boxcar integration, duration, duty, noise ablations. *Limit:* same Kepler source reused not independent stars or TESS confirmation.

P1-C07 Delay topology is omitted under the registered Stage 7 negative decision. *Control:* PSD preserving phase and IAAFT surrogates, classical features. *Limit:* no new Rudy phase claim.

P1-C08 Published identified modes support overlays and formula audits only. *Control:* Lund2017 modes and ratios. *Limit:* no blind degree inference or topological r01 r10.

References

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